

# CAN OGAN KARAGUN

Software Engineer

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Software engineer building production telecom systems and data-intensive services, with games shipped on Xbox and iOS. Experience in performance optimization, graph algorithms, distributed systems, graphics, physics simulation, and computer vision.

## EXPERIENCE

**Software Engineer - TechNarts, Ankara, Turkey | Sep 2025 - Present**

**STAR Suite - Network Management Platform for Turkcell**

- Selected from the existing STAR Suite team to join a 3-engineer platform modernization team in Feb 2026, rebuilding the system with Django, PostgreSQL, Redis, Kafka, and React.
- Reduced production-critical page response times from 20-30 seconds to 200-400 ms by eliminating N+1 queries, introducing snapshot-based read models, and adding Redis caching.
- Migrated telecom datasets containing hundreds of thousands to millions of records from MariaDB to redesigned PostgreSQL schemas using SQL migration pipelines.
- Developing graph-based fiber routing and network impact analysis to identify downstream effects of faults across telecom topologies.
- Consolidated Huawei, Nokia, and ZTE-specific logic into reusable backend services; built Huawei 400G inventory management covering equipment lifecycles and topology dependencies.
- Integrated vendor network management systems through REST, SOAP, and SSH-tunneled connections to synchronize inventory and performance data.

**Software Engineer Intern - KordSA | 2025**

- Developed Python tools for data processing and visualization to support analysis workflows.

**Software Engineer Intern - Kocaeli University IT Department | 2023**

- Implemented Redis caching for internal services.

## SELECTED PROJECTS

- [API Gateway](#) - Built a Flask reverse proxy with PostgreSQL-backed API keys, Redis rate limiting, retries, circuit breaking, and Prometheus metrics.
- [Redis Job Queue](#) - Built concurrent Go workers with Redis Streams, scheduled jobs, exponential-backoff retries, job status tracking, and a dead-letter queue; packaged with Docker Compose.
- [Go Distributed Microservices](#) - Built distributed backend services using Go, RabbitMQ, gRPC, and Docker.
- [3D Graphics Library / Rasterizer](#) - Built a software renderer from scratch in C, with matrix transformations, clipping, lighting, and textured triangle rasterization.
- [2D Physics Engine](#) - Built a physics engine in C++ with rigid-body simulation, collision handling, and constraints.
- [SLAM Mapping Simulation](#) - Built a robot simulation to map unknown environments using laser scans and Kalman filtering.
- [Real-Time Hand Tracking Game](#) - Built a Unity game controlled by real-time hand tracking using Python and OpenCV.

## PUBLISHED GAMES

Xbox: [Danger Close](#) and [Fight Freaks](#) (2021) - **70K+ combined downloads.**

iOS: [Encounter](#) and [Zigzag Chicken](#) (2022) - **35K+ combined downloads.**

## TECHNICAL SKILLS

**Languages:** Python, Go, SQL, JavaScript, C, C++, C#

**Frameworks & Data:** Django, Flask, Node.js, PostgreSQL, Redis, Kafka, RabbitMQ, REST APIs, gRPC

**Tools & Platforms:** Git, Linux, Docker, GitHub Actions, Unity, OpenCV

## EDUCATION

**Kocaeli University | B.Sc. Electronics & Telecommunications Engineering | May 2025**